TECHNICAL NOTE · NOVEMBER 2026

How Sailaware's routing engine finds its route

What the engine does when you start a routing, step by step: first the principles, with diagrams, then the exact implementation, setting by setting. Everything written here has been checked against the engine's code as of 11 November 2026.

- Isochrone method
- Runs on your computer
- Wind re-read every 5 minutes
- Straightened route, never slower

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PART ONE

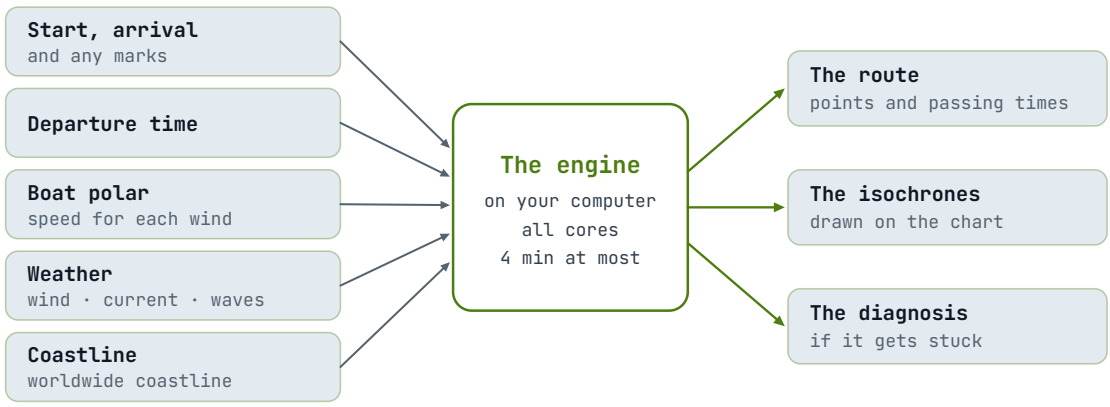
The principles

How the engine reasons, explained without formulas. Where useful, each section ends with "the precise term": the technical name, for those who want to look it up elsewhere.

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1 What the engine does

The question it answers is simple: **I leave from here, at this time, with this boat; which way should I go to arrive as early as possible?** The engine answers it with the forecast weather, without touching the coast, and leaving the marks on the right side.



What the engine receives: the course, the departure time, the polar, the weather and the coastline; what it returns: the route with its times, the isochrones, and a diagnosis if the computation gets stuck.

What it receives:

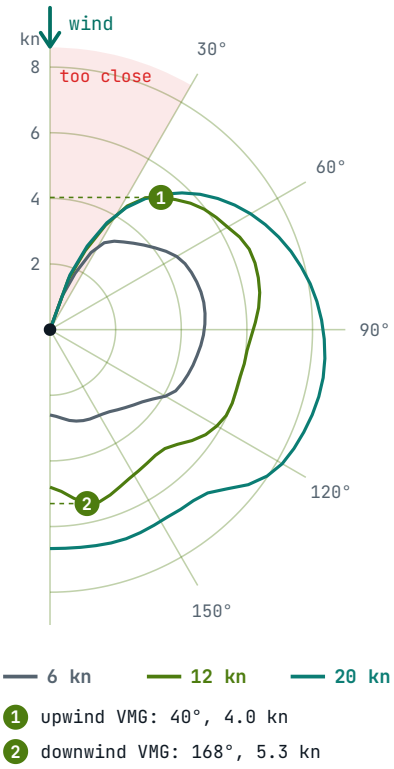
- **the start**: the boat's position (GPS, otherwise the estimated position, otherwise the last recorded point, otherwise the voyage's start), or a point chosen on the chart; **the arrival**; **the departure time**;
- **the marks**, if any: points to leave to port or to starboard, with a guard distance;
- **the polar** of the active boat;
- **the weather**: a wind file, required, and, if you have them, a current file and a wave file;
- **the coastline**;
- **the routing profile** chosen: Overview, Offshore, Standard, Coastal or Racing.

What it returns: the route, a sequence of points with the passing time at each one, and the arrival time; the isochrones, which the chart draws; and, if the computation does not arrive, the route up to the best point reached, with a diagnosis that says why and offers a new run.

The computation runs **on your computer**, on all the cores of its processor. Once the weather is downloaded, it depends on no server. It lasts four minutes at most. Run again with the same data, it returns the same route.

2 The boat: its polar

The polar says how fast the boat goes, depending on two things: **the wind strength**, and **the angle between its course and the wind**. The diagram shows the polar of the boat used in the Gulf of Morbihan test case.



A sailing boat's polar: its speed by true wind angle, for three wind strengths; upwind, the best speed towards the wind is reached at about 40° off the wind in 12 knots.

In 12 knots of wind, this boat makes 6.6 knots at 72° off the wind. At 40° off the wind it only makes 5.3 knots; at 24°, 2.6 knots. Head to wind, it does not move.

To go upwind, speed alone is not enough: what counts is **the speed towards the wind**, the VMG. At 40° off the wind, this boat makes 5.3 knots through the water and gains 4.0 knots towards the wind. Closer, it goes slower and gains less; further off, it goes faster but sails away. This is its **best upwind angle**. There is likewise a best downwind angle, here 168° off the wind. The engine always tries these angles (§ 6).

Between the values of the table

A polar is a table: here, one speed every 4° of angle and every 2 knots of wind. Between two angles, the engine draws a smooth curve; between two wind strengths, it takes the intermediate value. It never guesses beyond the table: in a wind stronger than its last column, it uses the last column.

The profile can apply an efficiency, for example 90% of the polar, and a separate efficiency at night. It can also count time lost at each tack and each gybe. By default, efficiency is 100% and no manoeuvre time is counted.

The precise term · speed polar, TWA (true wind angle), TWS (true wind speed), VMG (velocity made good).

3 The sea: wind, current, waves, coast

The wind

A wind file is a grid of points, for example every quarter of a degree, with a forecast every hour or every three hours: the **forecast times**. The engine uses a single file per computation, the one you have chosen.

The boat is almost never on a grid point, nor exactly at a forecast time. Between four grid points, the engine weights the four values by distance. **Between two forecast times, it follows the wind's strength and direction separately.**

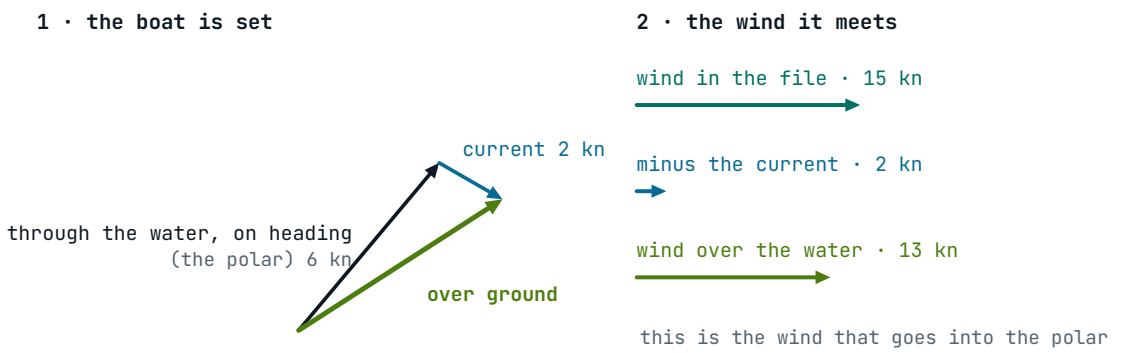
> **EXAMPLE**
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> The file gives 10 knots from 270° at noon and 20 knots from 300° at 15:00. At 13:00, a third of the way, the engine reads 13.3 knots from 280°: a wind that strengthens and veers at the same time, as it does in reality. A plain average of the two arrows would give 12.9 knots from 285°, a wind weakened while it turns.

One exception: when the wind turns by more than 150° between two forecast times, as when a front passes, there is no telling which way it turned. The engine then gradually returns to the average of the arrows, reached at 180°. Below half a knot, the wind is considered calm and its direction no longer counts.

Outside the grid or outside the period covered, the engine does not invent any wind: a boat that enters such an area stops. A start outside the file is refused before computing anything.

The current

If there is a current file, it acts twice. It **sets the boat**: its speed over ground is its speed through the water plus the current. And it **changes the wind the boat meets**: over moving water, the wind does not have the strength or direction given in the file. It is this wind over the water that goes into the polar.



Current: it is added to the boat's speed through the water to give its speed over ground; and it is subtracted from the wind in the file to give the wind the boat meets over the water.

Outside the area covered by the current file, the current counts as zero.

Waves

Waves do not change the boat's speed in this engine. They are only used as a limit: if you tick "Enable waves filter", a branch of the computation is dropped where the wave height exceeds the profile's limit, 6 m by default. The filter is off by default.

The coastline

The engine uses the worldwide coastline prepared by Sailaware from OpenStreetMap, at the level of detail of a zoom-12 map, once that file has been downloaded (Coastline data screen). Without it, it uses the GSHHS coastlines, if you have installed them. Without either, it does not test for land.

Every leg the engine considers is checked against the coast. It is refused **if it crosses the coast**, or **if one of its ends is less than 0.1 nautical mile from the coast** (the profile's margin, adjustable). One single exception: the last leg into the harbour may cross the coastline within its **last 0.6 nautical mile**, the time to enter a harbour drawn on the coastline or sheltered behind a breakwater.

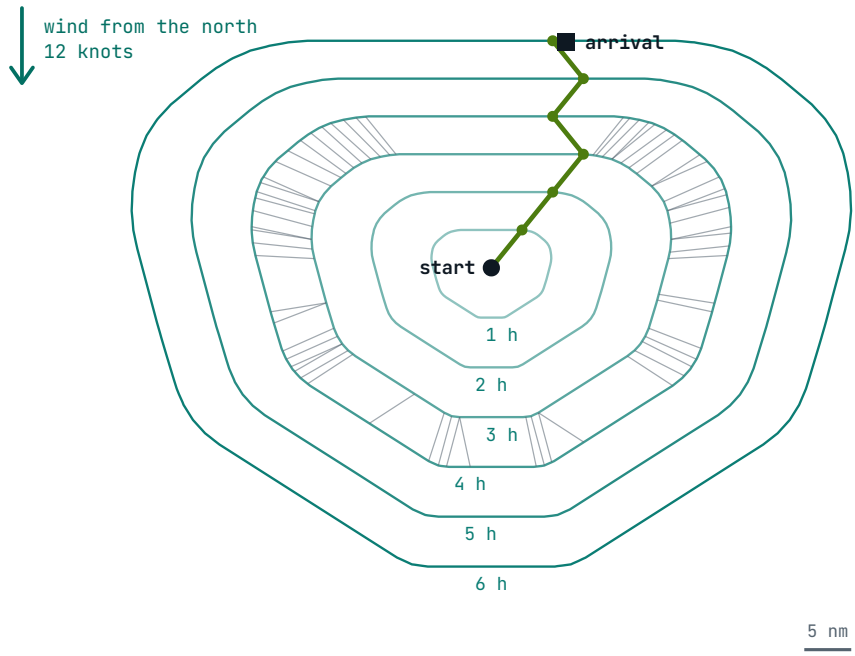
`The precise term · GRIB2 file, bilinear interpolation in space, polar interpolation (magnitude and angle) in time; surface current; significant wave height (Hs).`

4 The idea of isochrones

Imagine a fleet of boats identical to yours, leaving together from the same point, each in its own direction. After one hour, they draw a curve: all the points your boat can reach in one hour. This is an **isochrone**, literally "same time".

For the next hour, each boat on this curve sets off in turn in every direction. Of all the points reached, only those on the **outer edge** count: a boat left inside is behind its neighbours on the edge, and will stay behind. So the edge is kept, and the process starts again. Hour after hour, the curve grows like an oil stain, faster where the boat is fast.

The first moment the curve touches the arrival is the **earliest arrival time**. To know the route, each boat only has to remember the point it left from, its **parent**: from the arrival, you go back from parent to parent to the start.



Isochrones of a sailing boat in a steady wind: each curve joins the points reached in 1, 2, 3 hours; the route to the upwind arrival is read by going back through the parents; it tacks four times.

This drawing was made by a small separate program that applies the principle to the polar of § 2, in a steady 12-knot northerly wind. Upwind, the front flattens: the boat cannot sail straight into the wind, it gains ground by tacking at 40°. Downwind, it stretches. The upwind arrival is reached in 6 h 15. The route read back through the parents (in green) tacks four times: in a steady wind, these tacks are all roughly as good. The engine will regroup them at the end (§ 8).

`The precise term · isochrone method (Huygens' principle applied to navigation); wavefront; parent pointers.`

5 Why it is not that simple

The principle fits in three sentences. Doing it well, on real weather and a real coastline, raises four difficulties.

- **The number of boats explodes.** 100 points on the front, 30 headings each: 3,000 boats at the next step, then 90,000 if all are kept. At every step, only the edge of the front must be kept, and kept regular.
- **The wind changes during the step.** A step can last an hour: the wind is not the same at the beginning and the end of the step, nor at the start and the end of the leg.

- **The coast cuts the front.** Behind an island, in a bay, the front hollows out. An edge that is too "tight" would forget these pockets, where the fastest route often goes.
- **The announced time must hold.** A route drawn in broad strokes can promise a time the boat will not achieve. Each leg must be checked on a small scale.

The following sections show how the engine answers each of them.

6 One computation step, in order

The computation repeats the same sequence of operations, one isochrone after another, until the arrival, and at most 1,000 times. Here is one step, in the order the engine performs it.

1
Choose the step length
4 minutes near the start, 60 minutes offshore.

2
Launch the fans
From each front point, a series of headings outwards.

3
Move each boat forward
In 5-minute slices, with the wind re-read for each slice.

4
Drop the impossible
Land, wind beyond limits, mark on the wrong side, boats heading away from the goal.

5
Keep the outline
The outer edge, hollows included, with a bounded number of points.

6
Propose the arrival
Each point within reach of the harbour proposes its time; the earliest is kept.

Then everything starts again with the new isochrone. Steps 1 to 4 are detailed here; step 5, the heart of the engine, has its own section (§ 7), and so does step 6 (§ 8).

Step 1 · The step length

The step is not the same everywhere. Near the start and the arrival, where the coast and manoeuvres matter, it is short. Offshore, it is long, to go fast. The engine divides the trip into five zones, measured in hours of sailing at the boat's average speed:

start	moving away	plateau	approach	arrival
4 min	10 min	60 min	20 min	10 min
under 1 h	1 to 3 h	the rest	4 to 2 h	under 2 h

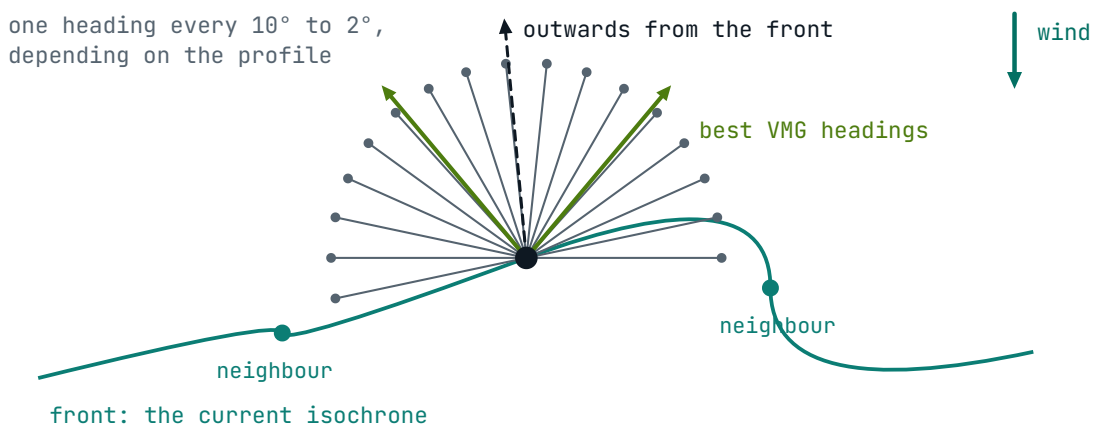
EXAMPLE

For the Gulf boat, the average speed of the polar is about 6 knots (the average from 30° to 180° off the wind and from 4 to 30 knots of wind). The start zone therefore covers roughly the first 6 nautical miles, the arrival zone the last 12.

The shortest step wins: as long as a single still-active boat of the front is in the start zone, the whole front advances by 4 minutes. The five profiles have the same steps. In the launch window, two time-precision sliders shorten or lengthen them, from a third to three times: one for the zones near the start and the arrival, the other for the plateau. The last step is cut exactly at the end of the forecast.

Step 2 · The fans

Each front point launches boats on a series of headings. These headings do not go all the way round: they open **outwards from the front**, between the directions of its two neighbours. This makes about half a turn, a bit more on a bulge of the front, a bit less in a hollow. Launching a boat inwards would serve no purpose: it would fall back behind the front.



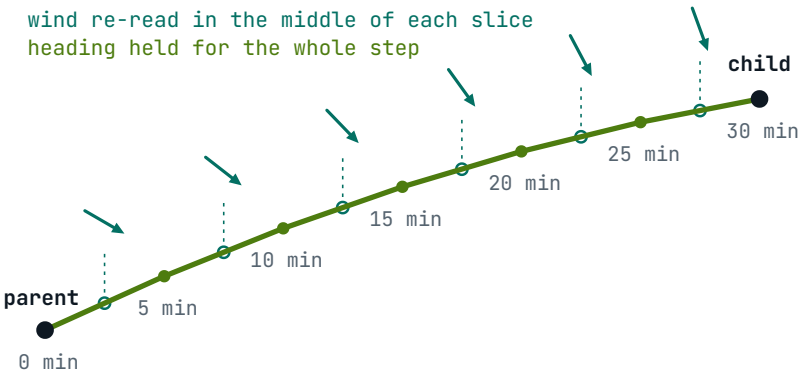
The fan of a front point: evenly spaced headings, opened outwards from the front between the directions of its two neighbours, plus the best VMG headings upwind and downwind.

The gap between two headings depends on the profile: **10° in Overview, 8° in Offshore, 6° in Standard, 4° in Coastal, 2° in Racing**. The four best-VMG headings are added, upwind and downwind, on both tacks, when they fall within the fan: they are often the best headings, and a 10° gap could miss them.

At the very first step, the start has no neighbours: it launches its boats all the way round. Along a coast, boats sometimes stay stuck in the same place for several isochrones in a row; their fan is then thinned out (one heading in ten). They keep every direction, with fewer headings, which avoids endlessly recomputing boats that no longer move.

Step 3 · Moving each boat forward

Each boat **holds its heading for the whole step**. But the engine does not compute its run in one go: it cuts the step into **5-minute slices** (a 60-minute step makes twelve). For each slice, it estimates where the boat will be in the middle of the slice, re-reads the wind and the current at that place and time, derives the speed from the polar, and moves the boat forward.



A 30-minute step cut into six 5-minute slices: the heading stays the same, the wind is re-read in the middle of each slice, where the boat then is.

Why the middle of the slice? Because it is the best estimate of the wind met during the slice. Reading the wind at the start of the step and keeping it for an hour distorts the arrival time as soon as the wind changes. On 120 runs with current, measured in October 2026, reading in the middle of each slice kept the actual duration within 0.13% of the promised duration, against deviations of up to 2.35% when reading at the start of the step.

A boat is abandoned if, on a single slice, it has no wind (outside the file), if the wind exceeds the profile's limit (70 knots by default, "Enable strong wind filter"), if the waves exceed the limit (filter on), or if the polar does not make it move.

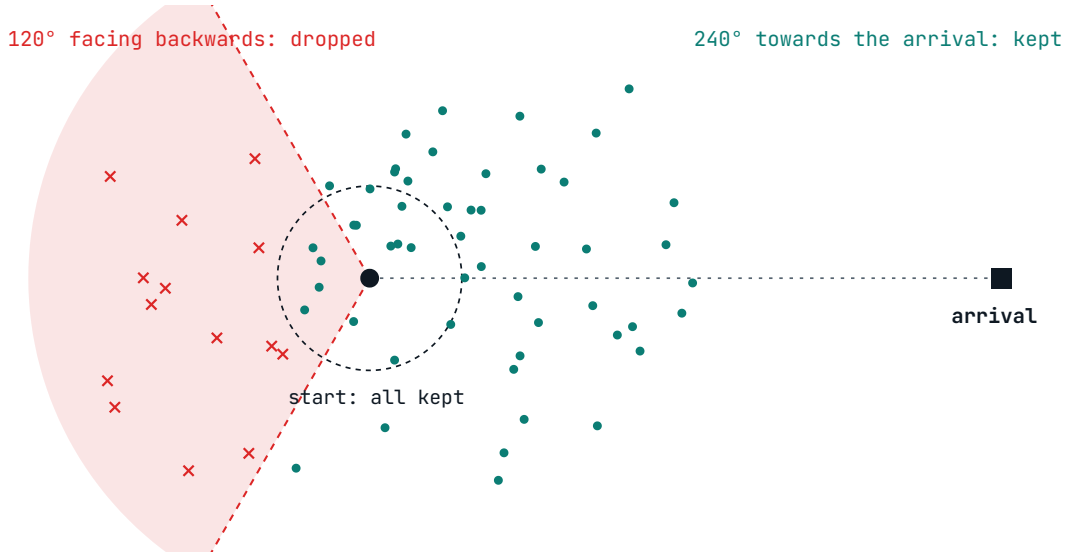
With "Hybrid" propulsion (Settings › Routing › Motoring), the boat switches to the engine, at 6 knots, when its best sailing speed towards the front's outside stays under 3 knots for more than 30 minutes, and it goes back to sail in the same way. By default, the profile is sail only.

Step 4 · Dropping the impossible

Land. Each leg, from the parent to the boat's new position, is checked against the coast (§ 3). With a current that changes during the step, the actual track is not straight: the coast is then tested slice by slice. To go fast, a boat whose parent is further from any coast than it can travel during the step, plus the margin and one mile, is not tested in detail: it cannot reach it.

Marks. A branch that passes a mark on the wrong side is removed (§ 9).

The cone. Boats heading clearly away from the arrival will lead nowhere. The engine keeps a cone facing the arrival, or the next mark, with its apex at the start; boats falling outside the cone are dropped. It opens to 240° in the five original profiles, and you can set it, in the profile or in the launch window: 120°, 180°, 240°, 300° or 360°. The tighter it is, the faster the computation, but a route that strays far from the axis may be missed; at 360°, nothing is dropped. Near the start (less than one hour of sailing) and near the arrival (less than two hours), everything is kept. And if the cone were to drop everything, it drops nothing.



The selection cone, here at 240°, its original setting: boats heading into the sector facing away from the arrival are dropped, except very close to the start. The cone can be set from 120° to 360°.

Crawling boats. Within 5 miles of a coast, the engine watches the lineages that no longer advance. Over two steps, a lineage must progress outwards from the front by at least 0.8 times the average gap between two front points, at each step; otherwise it is abandoned. This prevents boats stuck against a coast from cluttering the computation.

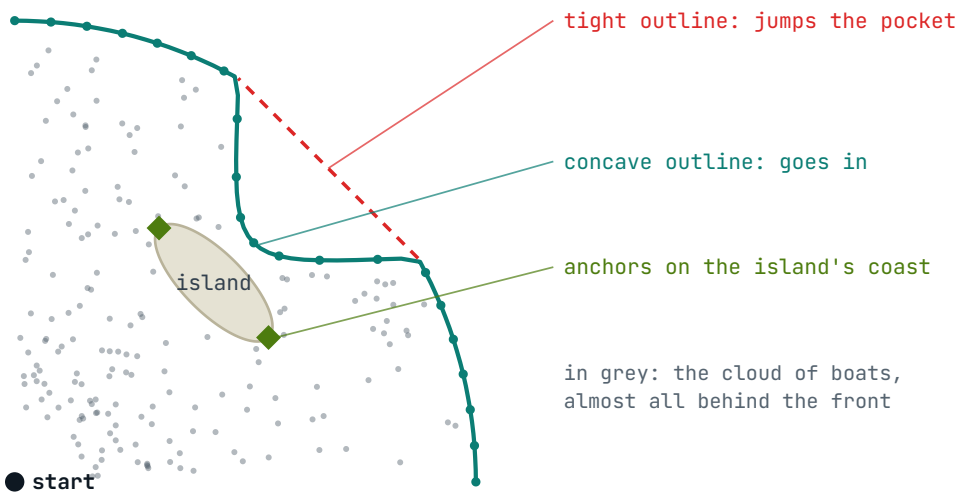
The precise term · adaptive time step by zones (SMART-DT); fan oriented on the front's normal; integration by 300 s sub-steps, midpoint reading; selection cone; anti-stagnation filter.

7 Keeping the right outline

After step 3, the engine has a cloud of thousands of points. It keeps only the outer edge, and the way it does so decides everything: this is where a route does, or does not, pass behind an island or deep into a bay.

An outline that keeps the hollows

The simplest outline would be the "tight" hull, like an elastic band around the points. It jumps over the pockets. The engine starts from that hull, then **digs** into it: a side that is too long, compared with the distance of the points lying just behind it, is broken in two and the outline goes down to those points. The two new sides are examined in turn, and so on, until no side can be dug any further. The profile sets how deep the digging goes: least in Overview, most in Racing.



Behind an island, a tight outline jumps over the pocket; the engine's concave outline goes into it, supported by anchors placed on the island's coast.

Not losing sight of the coast

Along a coast, the stuck boats disappear from the cloud. Without them, the outline would have nothing to lean on to go down into the pockets, and it would recede. The engine answers in two ways:

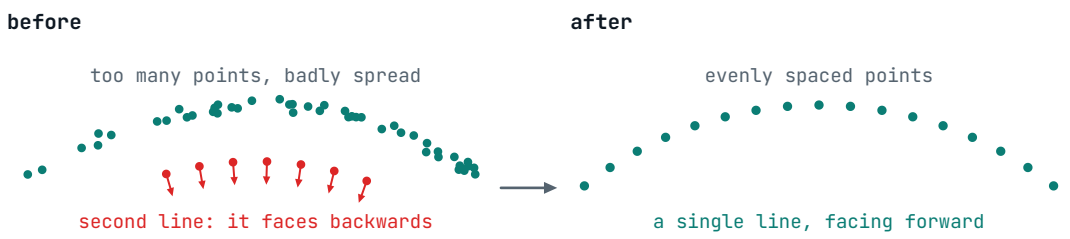
- it puts the points of the previous outline back into the cloud as **copies**: they no longer launch boats, they only hold the boundary;
- for each side of the previous outline longer than 10 miles, it places **two anchors on the nearest coast**. The outline can then follow the flank of an island or enter a bay, and the boats there become edge boats again, which will set off again.

A bounded number of points

The outline has a maximum number of points, proportional to its length: from 2 times a reference density in Overview to 6 times in Racing. Beyond it, points are removed by spreading them evenly along the outline, without touching the active boats that must be kept.

Removing what faces backwards

On an indented coast, the outline can fold back and form a second line, inside, whose points face backwards. Two checks remove it. First, each point is judged by the points of the previous isochrone that could have reached it during the step: if more than a third of them see it facing the wrong way, it is removed. Then, each point is compared with the nearest piece of the previous isochrone: if it faces the opposite way, it is removed.



Before: crowded, irregular points, and a second inner line whose points face backwards. After: evenly spaced points on a single line.

Spacing the points

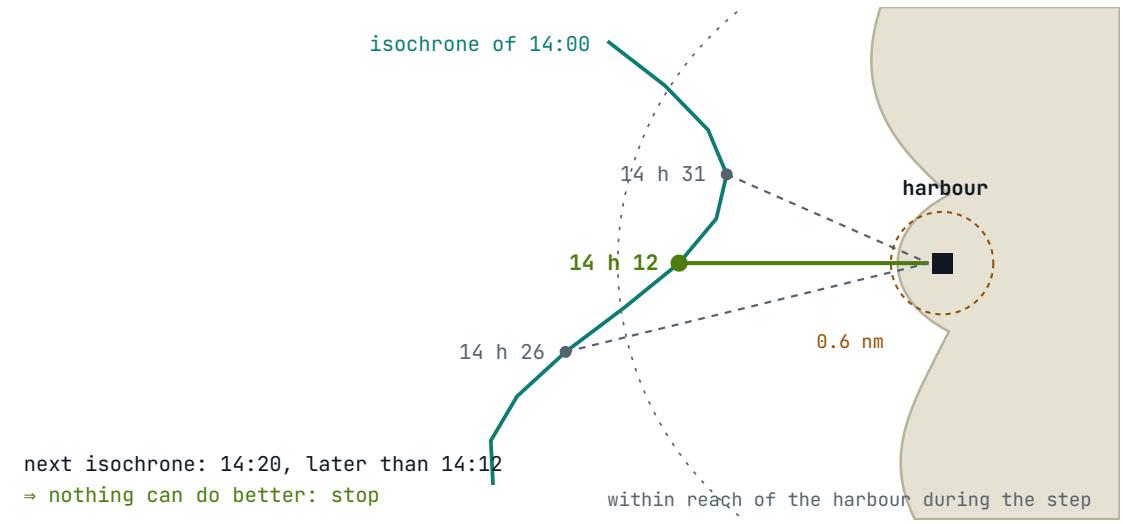
Finally, two points that are too close become one, the more advanced of the two, and a point lying on a straight line between its neighbours is removed. The copies of the previous outline are spared. If there are marks, a branch is never erased by a branch that has passed fewer marks.

The precise term · concave hull ("concaveman" algorithm); front memory; coast anchors; vertex cap; normal-opposition filters; regulation by decimation.

8 Arriving, then straightening

Earliest arrival

At each new isochrone, every point that can reach the harbour during the step, or that is less than one mile from it (the profile's "Arrival radius"), **proposes an arrival time**. The engine sails this last leg like the others, in 5-minute slices, with the heading corrected for current. The leg is refused if it crosses the coast before the last 0.6 mile, or if it arrives after the end of the forecast. The earliest proposal is kept.



Arrival: every point of the front that can reach the harbour during the step proposes its time; the earliest is kept, and the computation stops as soon as the next isochrone would fall after it.

The computation stops **as soon as the next isochrone would fall after the best proposal**: no boat could do better any more. A safety net completes the rule: if no point proposes an arrival while a side of the outline passes within reach of the harbour, the two ends of that side propose.

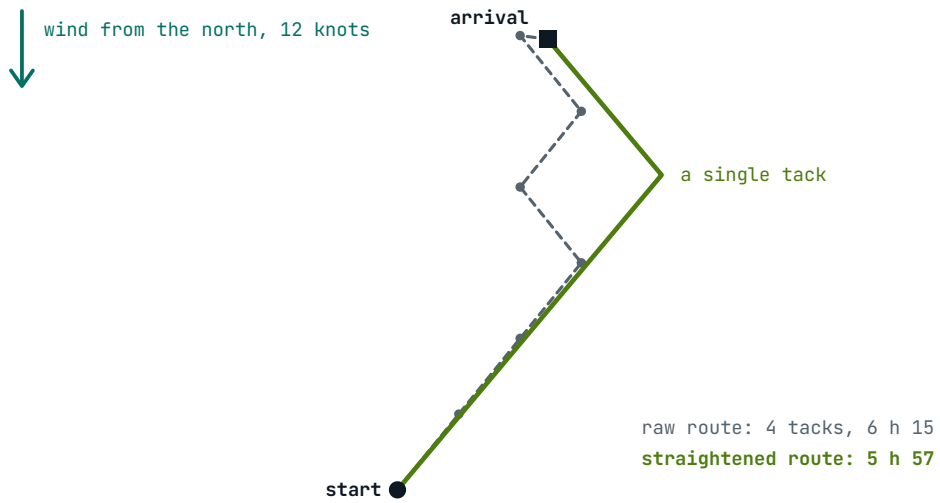
The raw route

The route is read back from parent to parent, then a last straight leg joins the harbour. This raw route is correct, but it meanders: at points of sail where a single heading would do, it changes heading slightly at each isochrone, and it often tacks or gybes much more than needed. On real routes, 5 to 38 manoeuvres were counted where a single tack was enough.

Straightening

A pass re-reads the route after the computation, without changing anything in the computation itself. It does three things.

1. **Straighten.** From each point, it looks for the longest straight leg to a point further along the route, provided it gets there no later than the original route.
2. **Regroup the tacks.** A sequence of tacks or gybes is replaced by longer boards, at the same angles, without leaving the corridor of the original route.
3. **Leave the marks alone.** Legs that pass a mark stay as they are.



The raw route tacks four times; straightened, it tacks only once and arrives earlier. The pass never keeps a slower route.

The golden rule is **"never slower"**: the new route is timed from end to end, and it is kept only if it arrives no later than the original route. Otherwise, the engine tries the route that is only straightened, without regrouping; otherwise, it returns the original route unchanged.

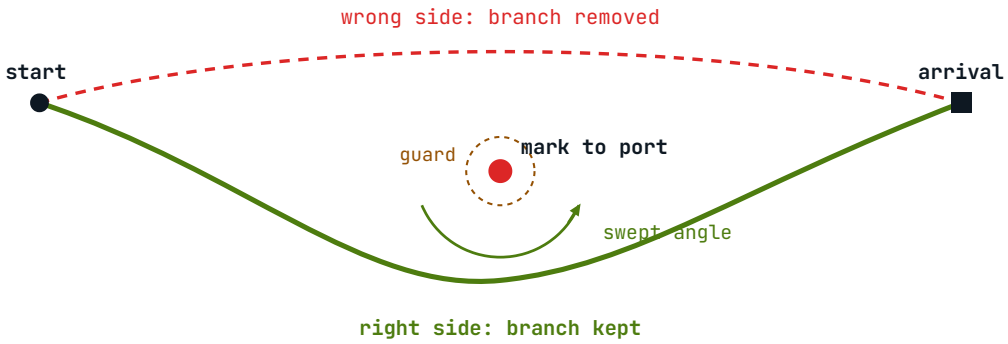
The stopwatch

Each new leg is sailed in 5-minute slices, at the time the boat will be there. Every 60 seconds of sailing, the engine also checks that the boat can really hold this straight line: a cross current can make it untenable over its last metres. **The times shown on the route are those of this stopwatch.**

The precise term · arrival by proposals (earliest arrival); reconstruction by pointers; "never slower" shortcut pass; timing by sub-steps.

9 Marks

A mark is a point to leave on one side, port or starboard, at a guard distance. The engine does not draw a detour around it in advance: it lets the computation find the best way to pass it, and it judges each branch along the way.



A mark to leave to port: the branch that goes round the mark in the right direction is kept, the one that passes on the wrong side is removed; a guard distance keeps the boat clear.

- Each branch measures **the angle it has travelled around each mark**, with its direction. A mark is passed when the branch has gone at least halfway round the required turn.
- A branch turning the wrong way is removed. A leg passing closer than the guard distance (1 metre by default, adjustable) removes the branch for good.
- When the outline is kept, a branch is never erased by a branch that has passed fewer marks. The cone recentres: its apex moves to the last mark passed, and it aims at the next one.
- A route has only arrived if every mark has been passed on the right side, last leg included. The application also re-judges the returned route with the same rules.

If a mark cannot be held, for example because it lies on land, the computation does not run forever: once a branch has touched the arrival, if for two hours of sailing the best branch gains neither 5% of the turn around its next mark nor one mile towards it, the computation stops and names that mark. Up to 20 marks can be set.

`The precise term · signed angular tracking per mark; sorting by progress; guard distance; "untenable mark" stop.`

10 When the computation gets stuck

The computation's limits

The computation stops after **4 minutes**, or if it exceeds **6 GB of memory**, or after **1,000 isochrones**, or at the end of the forecast. If it already has an arrival proposal at that moment, it returns it as an arrival. It also stops when the front is empty, because every boat is blocked: if at least half of them were stopped for lack of wind, the forecast is exhausted or the route leaves the weather area, and the engine returns the route so far.

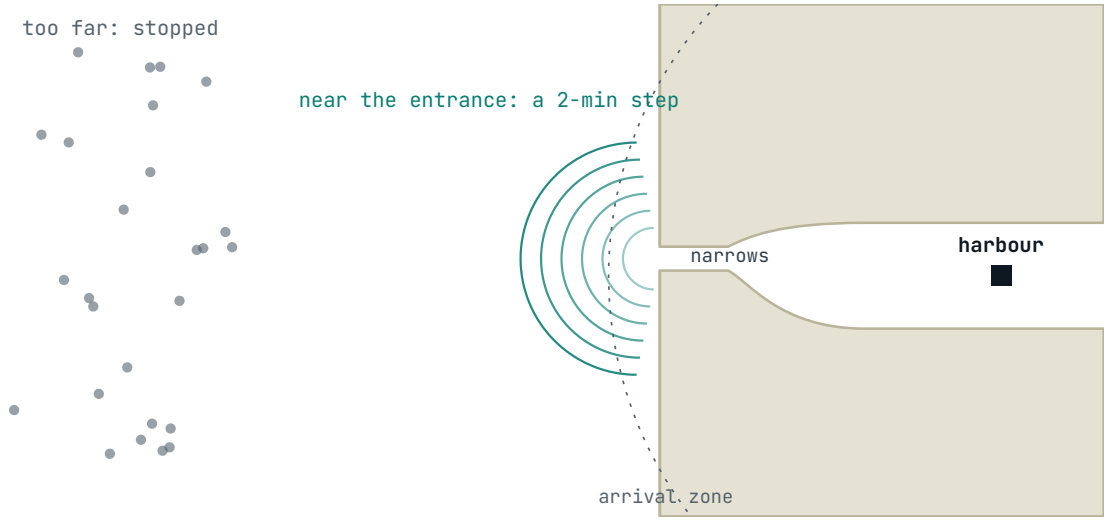
The best boat

If the computation stops without arriving, the route goes to the **best boat of the whole computation**: the one that has passed the most marks, then the closest to the arrival. The rest of the way is drawn as a straight line, without times.

A rule also avoids waiting for nothing. If a boat has touched the arrival zone and all the still-active boats then move away from it beyond 96 hours of sailing, the computation stops: it will no longer arrive. Without this rule, a Gulf of Morbihan test case simulated 358 hours of sailing instead of 154.

The narrow-passage rescue

Narrows, the entrance of a gulf, a channel between two islands can escape the computation: few boats get in, and the step is too long to slip through. If the computation stops without arriving, but its route ends near the arrival with all its marks passed, the engine **runs again once**, in the time left out of the 4 minutes.



The narrow-passage rescue: boats that went too far are stopped and the step is tightened near the entrance, so that the computation finds the narrows.

During this new run, as soon as a boat that is up to date with its marks touches the arrival zone, the boats that went far away (beyond three to four hours of sailing from the harbour) stop advancing, and the step near the arrival drops to **2 minutes**. The computation thus concentrates on the entrance. The new run is kept only if it arrives. Live routing, which runs on its own while sailing, does not do it.

The diagnosis

When a route does not arrive, or arrives with a flaw, the diagnosis screen says what happened: no passage found, all boats blocked, an untenable mark, a start outside the weather, an unreadable file, the end of the forecast, leaving the weather area, time or memory exhausted, or a last leg crossing more than 50 m of land. It offers to run again with the profile **just more precise** (when no passage was found) or **just faster** (when time or memory ran out), to move the arrival, to open the weather, or to edit the marks.

11 The five profiles

Each profile sets the fineness of the computation. The finer it is, the closer the route gets to the best possible one, and the longer the computation takes.

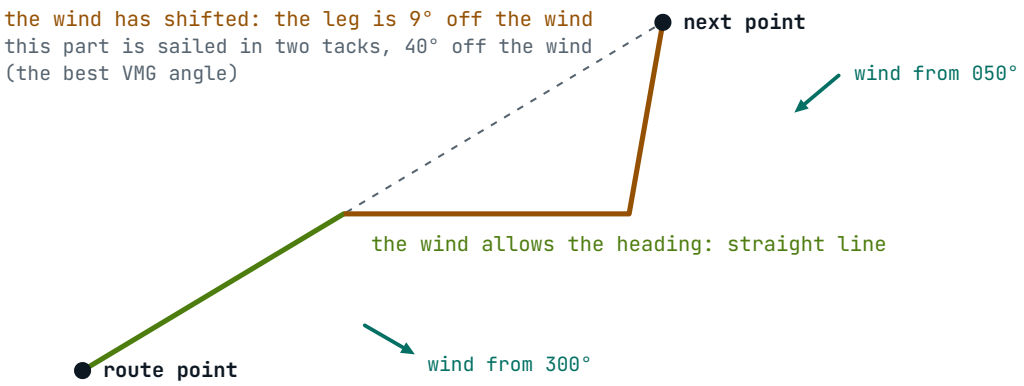
PROFILE	GAP BETWEEN HEADINGS	OUTLINE DIGGING	OUTLINE POINTS
Overview · quick	10°	shallowest	×2
Offshore · open sea	8°	shallowest	×3
Standard · balanced	6°	medium	×4
Coastal · coastal	4°	deep	×5
Racing · precise	2°	deepest	×6

The five original profiles share the rest: the time steps, the 240° cone, the coast margin, the wind and wave limits, the maximum computation time. You can edit your profiles in Settings › Routing; the values above are the original ones. The detailed figures are in § 22.

12 Re-timing a route

A route already drawn, computed by Sailaware or imported from another program, can be re-timed with other weather. The engine does not look for a new route: it **replays this one**, with the same physics and the same wind reading. For a Sailaware route, it follows the clock of the engine that computed it. For an imported route, it sails each leg in a straight line, in 5-minute slices.

When a leg comes too close to the wind to be held, that part is sailed **in two tacks, at the best VMG angle**. In Simple track, the chart shows it in amber, only on the part sailed in tacks, and in red the legs where the boat makes less than 3 knots. The route table gives the heading of both tacks. Tacking time is not counted there.



Re-timing a straight leg: as long as the wind allows the heading, the boat follows it; when the leg comes too close to the wind, that part is sailed in two tacks, at the best VMG angle.

If the boat can no longer move forward, the re-timing stops and says why: head to wind, wind too strong or too light, sea too rough, end of the forecast, leaving the weather area, current stronger than the boat. A separate search reports the legs that cross 50 m of land or more.

While sailing, live routing runs a complete computation, silently, ten minutes after the end of the previous one, from the boat's position. Between two computations, a light follow-up updates the arrival time every 15 seconds.

PART TWO

The implementation

The same mechanisms, this time with their exact values, their formulas and their order in the code. For those who want to understand the algorithm all the way, or compare it with another.

---

13 Where the computation runs

The engine is written in Rust and is part of the application, on macOS as on Windows. The computation is spread over all the cores: each front point computes its fan in parallel. The results are then put back in a fixed order, and the internal tables have no random order. The same computation therefore returns the same route, bit for bit, on the same machine; the test benches check this by comparing each route with a recorded snapshot, system by system. The only exception: a computation cut by the 4-minute limit depends on the machine's speed.

Weather files stay in memory from one computation to the next (8 files or 512 MB at most), and a file is loaded as soon as you activate it, if it is not too large. The coastline too (256 tiles or 512 MB at most). During the computation, each isochrone is sent to the screen, reduced to 400 points for display, and you can stop the computation at any time.

Before computing, the engine refuses an inconsistent request: a start outside the period or area of the wind file, a cone outside 120° to 360°, badly placed marks.

14 The complete loop

Here is the main loop, written in plain words. Each line refers to a section of this note.

```
front ← { start }                          isochrone 0, at the departure time
best ← arrival proposal from the start          (without marks)
for k = 1 … 1,000:
    if 4 min elapsed or memory > 6 GB: exit             § 19
    Δt ← step of the finest zone holding an active boat    § 16
    if time(front) + Δt ≥ time(best): return best        § 19
    cut Δt at the end of the forecast
    cloud ← ∅
    for each point P of the front, in parallel:
        for each heading c of P's fan:                  § 6, step 2
            boat ← P
            for each 5-min slice of the step:             § 15
                read wind, current, waves mid-slice
                speed ← polar(wind over water) on heading c, + current
                if impossible: abandon this boat
                move the boat forward
            if leg P → boat touches the coast or its margin: abandon  § 18
            add (boat, parent P, heading c) to the cloud
    remove: wrong side of a mark, outside the cone, crawling lineages
    front ← outline(cloud)                                  § 17
    for each front point within reach of the harbour:
        sail the last leg; best ← the earliest             § 19
    if front empty, mark untenable, fleet gone away: exit
route ← go back through the parents from best
route ← straighten, regroup, time                     § 20
```

At rank 1, the whole cloud forms the front, without an outline: the start has neither neighbours nor hollows yet. If the computation exits without a proposal, the route goes to the best boat of the computation, and the rescue run may be played (§ 19).

15 Reading the weather and the polar

Wind

- **Format**: GRIB2 only, on a regular latitude-longitude grid (other grids are refused). Packings read: simple, PNG, JPEG 2000, CCSDS. The engine takes the U and V wind components and converts them to knots.
- **In time**: between the two forecast times around the instant, each grid cell is first blended in **magnitude and angle**, each linearly, the angle along the shortest way. If the angle turns by more than 150°, the blend gradually slides towards blending the U and V components, reached at 180°. Below 0.505 knot, a forecast time is considered calm and its direction is ignored.
- **In space**: bilinear interpolation of the components, between the four cells around the point.
- **When**: in the middle of each slice of at most 300 s, at the estimated mid-slice position (advanced at the previous slice's speed).
- **Never any extrapolation**: outside the grid or the file's period, there is no wind.

Current

- Separate, optional GRIB2 file, active as soon as it is provided; read like the wind, with a calm threshold of 0.05 knot.
- Speed over ground = speed through the water on the heading + current vector. The heading and the wind angle remain those of the boat through the water.
- Wind going into the polar = wind in the file – current (the true wind over the water).
- Outside the file's area: zero current.

Waves

Only the significant height Hs is read (encoded as in GFS wave files), bilinearly then linearly in time. It is only used for the profile's limit, and only if the filter is on.

Polar

- **Along the angle**: cubic Hermite curve for each wind column, with Catmull-Rom tangents, extended to 0° and 180°.
- **Along the wind strength**: linear interpolation between two columns.
- **At the edges**: below the smallest column, the smallest; above the largest, the largest; speed capped at 1.1 times the column's maximum; port and starboard symmetrical.
- **Efficiencies**: overall and at night, as percentages; night starts when the sun goes 0.833° below the horizon. They apply only if they differ from 100%.
- **Best-VMG angles**: searched degree by degree in the polar, for the wind at each point.

Geometry

Positions are computed in the Mercator plane, where a constant heading is a straight line; the distances that matter for navigation (reaches, radii, margins) are converted to true nautical miles. The reference meridian is recentred on the trip, which allows crossing the antimeridian.

16 The time step

ZONE	WHERE	STEP
start	less than 1 h × $\bar{v}$ from the start	4 min
moving away	1 h to 3 h × $\bar{v}$ from the start	10 min
plateau	everywhere else	60 min
approach	4 h to 2 h × $\bar{v}$ from the arrival	20 min
arrival	less than 2 h × $\bar{v}$ from the arrival	10 min

- $\bar{v}$ is the average speed of the polar, from 30° to 180° off the wind and from 4 to 30 knots of wind.

- The step of an isochrone is the finest of the steps of the zones holding its active boats; copies and blocked boats do not count. A branch that has passed a mark is no longer "at the start".
- Two factors from the launch window multiply the steps, in 7 notches from 1/3 to 3: one for the four near zones, the other for the plateau.
- The last step is cut at the end of the forecast; an isochrone placed less than one minute before that end is the last one.
- During the rescue run, the arrival zone's step drops to 2 min.

17 Building the front

The engine first measures the front's **resolution**: the average progress of a step multiplied by the gap between two headings (in radians), smoothed when it drops. It is the expected gap between two neighbouring points; several thresholds are expressed in resolutions. Then, from rank 2 on, the following chain runs, in this order.

1. **Copies of the previous outline**: its points are put back into the cloud, sterile (they do not propagate), and protected from decimation if kept.
2. **Grid sorting**: cells of 2.0 / 1.5 / 1.25 / 1.25 / 1.25 resolutions (from Overview to Racing); in each cell, the point furthest from the cloud's centre is kept.
3. **Concave hull** ("concaveman" algorithm): start from the convex hull; for each side AB longer than the threshold L, look for the inner point closest to the segment, and closer to this side than to the neighbouring sides; dig (A → p → B) if |AB| divided by the distance from p to the nearer of the two ends reaches the **concavity**: 1.5 (Overview, Offshore, Standard), 1.25 (Coastal), 1.0 (Racing). The two new sides go back into the queue.
   The threshold L is 1/ρ, where ρ = density × $(100 / R)^{0.5}$ points per nautical mile, R being the front's mean radius in nautical miles, and the density being 0.75 / 0.75 / 1.0 / 1.25 / 1.5.
4. **Reinsertion** of the points left under chords longer than 100 resolutions.
5. **Cap**: N_{max} = ρ × perimeter × margin, with a margin of 2 / 3 / 4 / 5 / 6; evenly spaced decimation along the outline, protecting live points.
6. **Coast anchors**: for each side of the previous outline longer than 10 nautical miles, its two ends are projected onto the nearest coast and injected as sterile points; a second digging pass leans on them.
7. **Unification**: each point q of the previous isochrone, with reach R_{max} = maximum polar speed × Δt, examines the points p of the new outline lying in its forward half-plane (180°) and within reach; p is opposed if its normal and that of q are more than 90° apart. p is removed if more than a third of its examiners find it opposed; a point without examiners is kept.
8. **Normal opposition**: each point is projected onto the nearest edge of the previous isochrone; it is removed if its normal is opposite to that edge's (one pass).
9. **Regulation**: proximity decimation at 0.3 resolution (the more advanced point is kept), then removal of aligned points (threshold 0.1 × the decimation distance); copies are protected.
10. **Sorting by progress**, only when there are marks.

The thinning of blocked coastal boats (§ 6) targets clusters of boats that have stayed in place for four isochrones in a row near a coast: their gap between headings is multiplied by 10 at the next isochrone.

Anti-stagnation concerns lineages whose parent is less than 5 miles from a coast: their progress, projected onto the normal frozen at the start of monitoring, must reach 0.8 resolution per step, accumulated over two steps, otherwise the lineage is rejected.

Other mechanisms were tried during development and remain in the code, switched off: local domination, bisection refinement of headings, retrieval, freezing of coastal boats, alpha-shape filter.

18 The coastline test

- **OpenStreetMap coastline**: zoom-12 vector tiles, prepared by the Sailaware service, used only if the whole world is present at that zoom; read in 10° × 10° squares (with one tile of margin); lines lying on tile borders, which are only cuts, are ignored; latitudes up to ±85.06°.
- **GSHHS**, otherwise: resolution of your choice ("h" by default), coastline and Antarctic levels only: lakes are not read. Spatial index per 10° tile.
- **The test**: a leg is refused if it crosses a coastline edge (computed in the longitude-latitude plane), or if one of its two ends is closer than the margin to an edge (0.1 nautical mile by default). With a changing current, each slice is tested.
- **Quick sort**: if the parent is further from the coast than 1.01 × maximum speed × Δt + margin + 1 nautical mile, its legs are not tested (except under engine).
- **Per-boat cache**: nearby edges are gathered once, within a radius of (maximum speed + current) × Δt × 1.25 + 1 nautical mile; the verdict is identical to the full test.
- **Harbour entry**: the last leg may cross the coastline within the last 0.6 nautical mile, without margin; the length of land crossed is measured, and the screen reports it from 50 m.

19 Arrival and stops

Proposals: at each isochrone, every point less than maximum speed × Δt from the harbour, or within the arrival radius (1 nautical mile), sails the last leg in 5-min slices, heading corrected for current. Refusals: land before the last 0.6 nautical mile, arrival after the end of the forecast, mark not held. Safety net: failing any proposal, the two ends of each side of the outline that passes within reach propose. At the start, without marks, a proposal is evaluated right away.

END OF COMPUTATION	WHEN	ROUTE RETURNED
arrival	the next isochrone would fall after the best proposal	the best proposal
time	4 min of computation	the best proposal if there is one, otherwise the best boat
memory	6 GB	same
isochrones	1,000 isochrones	same
forecast exhausted	less than 60 s before the end of the file	same

END OF COMPUTATION	WHEN	ROUTE RETURNED
empty front	all boats blocked	partial if at least half for lack of wind, otherwise none
untenable mark	2 h without gain on the next mark	the best boat
fleet gone away	all active boats more than 96 h × v̄ from the harbour, after a boat has touched the arrival zone	the best boat ("no passage")

The best boat: the most marks passed, then the closest to the target; the returned route goes to it, followed by a remainder in a straight line, without times.

The rescue run: only for a routing started by hand; once only; if the computation did not arrive without hitting the time or memory limit, and the end of its route is in the arrival zone (2 h × v̄ from the harbour) with all its marks. As soon as a boat that is up to date with its marks touches this zone, the boats beyond max(1.5 × 2 h × the boat's maximum speed, 2 zone radii) stop propagating, and the arrival zone's step drops to 120 s. Budget: what is left of the 4 minutes. The run is kept only if it arrives.

20 The straightening pass

1. **Straighten**: from each kept point, the longest straight leg to a point further along the original route, up to 128 points further, searched by doubling jumps (2, 4, 8… points) then by bisection; it must get there no later than that point's reference time. Where no shortcut starts, the original segment is kept.
2. **Regroup the tacks**: a sequence of alternating boards, up to 64, is replaced by longer boards, at the board angles of the original route or at the best-VMG angles, going from one side of the corridor to the other; the corridor is the band of the original points around their chord. A sequence that cannot be regrouped in one go is split in two, up to 5 times.
3. **Frozen segments**: those that pass a mark, and those sailed under engine. Each new leg keeps the guard of every mark and sweeps the same angle around each one; the whole route is then judged again.
4. **Never slower**: the new route must arrive no later than the earlier of two times: the one announced by the engine, and the stopwatch replayed on the original route. Otherwise the route that is only straightened; otherwise the original route.
5. **Stopwatch**: each new leg sailed in 5-min slices at the boat's time; every 60 s of sailing, the point of sail is re-read with the current of the moment, to check that the boat holds the straight line; the coast is judged on the new legs only. The times of the returned points are those of the stopwatch.

21 Marks, in detail

- Each branch carries, for each mark k, the signed angle Θ_k swept around the mark since the start, in the Mercator plane. The course plan (start → marks → arrival) sets, for each mark, the angle to travel on the required side, x_k.

- Mark k is passed when the angle travelled, counted in the required direction, reaches x_k / 2. A branch that turns too far the wrong way (a quarter to half a turn depending on the mark) is removed.
- At the arrival: all marks passed, and no extra turn around any of them, last leg included.
- Guard: a straight segment closer than the guard distance to a mark, active or not, makes the branch "too close" for good.
- Sorting by progress in building the front; a passing line for each mark, which allows dropping branches that lag behind; cone recentred on the last mark passed.
- "Untenable mark" stop: from the first isochrone that touches the arrival, as long as no branch holds every mark, if the best one gains neither 5% of the half-turn of its next mark nor 1 nautical mile towards it in 2 h of sailing.
- At most 20 marks.

22 All the settings

What changes with the profile

PROFILE	GAP BETWEEN HEADINGS	CONCAVITY	DENSITY	SORTING CELL	CAP
Overview	10°	1.5	0.75	2.0	×2
Offshore	8°	1.5	0.75	1.5	×3
Standard	6°	1.5	1.0	1.25	×4
Coastal	4°	1.25	1.25	1.25	×5
Racing	2°	1.0	1.5	1.25	×6

What the five share

SETTING	ORIGINAL VALUE
time steps	4 / 10 / 60 / 20 / 10 min; zones of 1 h, 3 h, 4 h, 2 h
sailing slice	5 min (300 s), wind read mid-slice
selection cone	240°; adjustable to 120°, 180°, 240°, 300° or 360°
maximum duration	4 min
maximum memory	6 GB
isochrones at most	1,000
arrival radius	1 nautical mile
land tolerated at harbour entry	0.6 nautical mile

SETTING	ORIGINAL VALUE
coast margin	0.1 nautical mile
maximum wind	70 knots (filter on)
maximum waves	6 m (filter off)
anti-stagnation	within 5 nautical miles of a coast, 0.8 resolution per step
propulsion	sail only; in hybrid: 6 knots under engine, below 3 knots for 30 min
manoeuvres	no time counted
polar efficiency	100%, day and night
"holds the line" check	every 60 s of sailing
mark guard	1 m

23 What the engine does not do

- It uses only **one wind model** per computation; it does not blend several models.
- **Waves do not slow** the boat; they are only used for a limit, off by default.
- **Manoeuvre time** is not counted by default, and it never is in the tacks of a re-timing.
- The **coast margin** is checked at the ends of each leg or each slice (at most 5 minutes of sailing); in between, only crossing the coast is forbidden.
- Outside the current file's area, the **current counts as zero**, without warning.
- With GSHHS coastlines, **lakes** are unknown.
- The polar is **never extrapolated**, nor is the weather: no route beyond the wind's last forecast time.
- A **mark to round by less than 90°** is not handled yet.
- A computation lasts **4 minutes** at most; a very long trip in Racing may not fit in that time, and the diagnosis then suggests a faster profile.

Glossary

isochrone	The line of all the points the boat can reach at the same instant.
front	The isochrone being computed, from which the boats of the next step set off.
step	The time between two isochrones: from 4 to 60 minutes.
slice	A 5-minute piece of a step, over which the wind is re-read.
fan	The series of headings launched from a front point.
parent	The point a boat left from; the route is read from parent to parent.

branch, lineage	The sequence of positions of a virtual boat and its ancestors.
concave outline	The edge of the cloud of points, which follows the hollows instead of stepping over them.
resolution	The expected gap between two neighbouring points of the front.
polar	The table of the boat's speeds, by wind strength and wind angle.
true wind angle (TWA)	The angle between the boat's heading and the direction the wind comes from.
VMG	The speed towards the wind (upwind) or away from it (downwind).
forecast time	One forecast hour in a weather file.
GRIB	The format of weather forecast files.
mark	A point to leave to port or to starboard.
guard	The minimum distance at which a mark is passed.
cone	The sector facing the goal (240° by default, adjustable), outside which boats are dropped.